

2 (a) change in velocity / time (taken) **or** rate of change of velocity B1

(b) (i) $v_x = (24 / 1.5) = 16 \text{ (ms}^{-1}\text{)}$ A1

(ii) $\tan 28^\circ = v_y / v_x$ **or** $v_x = v \cos 28^\circ$ **and** $v_y = v \sin 28^\circ$ C1

$v_y = 16 \tan 28^\circ$ **or** $v_y = 16 \times (\sin 28^\circ / \cos 28^\circ)$ **so** $v_y = 8.5 \text{ (ms}^{-1}\text{)}$ A1

(iii) $v = u + at$ C1

$t = (0 - 8.5) / (-9.81)$
 $= 0.87 \text{ (s)}$ A1

(iv) straight line from positive v_y at $t = 0$ to negative v_y at $t = 1.5 \text{ s}$ M1

line starts at (0, 8.5) and crosses t -axis at (0.87, 0) and does not go beyond $t = 1.5 \text{ s}$. A1

(c) (i) $(v^2 = u^2 + 2as)$ $0 = 8.5^2 + 2(-9.81)s$
or $(s = ut + \frac{1}{2}at^2)$ $s = 8.5 \times 0.87 + \frac{1}{2} \times (-9.81) \times 0.87^2$
or $(s = vt - \frac{1}{2}at^2)$ $s = 0 - \frac{1}{2} \times (-9.81) \times 0.87^2$
or $(s = \frac{1}{2}(u + v)t$ **or** area under graph) $s = 0.5 \times 8.5 \times 0.87$ C1

$s = 3.7 \text{ (m)}$ A1

(ii) $\Delta E_p = mg\Delta h$ (allow $E = mgh$) C1

$m = 22 / (9.81 \times 3.7)$
 $= 0.61 \text{ (kg)}$ A1

(d) acceleration (of freefall) is unchanged / not dependent on mass, and so no effect (on maximum height)

or explanation in terms of energy:

(initial) $\text{KE} \propto \text{mass}$, $(\Delta)\text{KE} = (\Delta)\text{PE}$, (max) $\text{PE} \propto \text{mass}$, and so
no effect (on maximum height)

B1